

Research paper

Vibration-based distribution transformer fault prognosis using recurrent neural networks with self-attention

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ABSTRACT

Distribution and power transformers are crucial equipment in AC energy systems, serving purposes such as voltage conversion, voltage regulation, load balancing under imbalanced conditions, and even harmonic elimination. Reliable monitoring of these equipment is essential for grid stability. In recent years, vibration analysis has emerged as an effective method for assessing transformer condition. Deep recurrent neural networks (RNNs), specifically bidirectional long short-term memory (BiLSTM) and bidirectional gated recurrent unit (BiGRU) architectures, have already shown promising results in analyzing vibration signals for transformer fault prognosis. In parallel with these achievements, another innovation, known as the multi-head self-attention (MHSA) mechanism, has gained interest due to its effectiveness in analyzing sequential data. However, it remains unclear whether MHSA can enhance RNN performance for transformer fault prognosis—an issue that this study addresses. As the outcome of this exploration, the MHSA-BiGRU model achieves a 48% reduction in relative absolute error (RAE) compared to BiGRU when predicting excitation voltage. Similarly, the MHSA-BiLSTM model reduces RAE by 28% compared to BiLSTM for interturn short-circuit current prediction. These results demonstrate that integrating MHSA improves RNN-based models for transformer fault prognosis using vibration signals, offering a promising direction for more accurate and robust condition monitoring in power systems.

1. Introduction

Risk assessment plays a crucial role in many industries, especially those that involve the use of expensive equipment. Businesses and large companies employ various instruments for early equipment fault detection to avoid the potential economic costs associated with equipment malfunctions. To quickly identify anomalies associated with stable equipment operation, engineers strategically deploy sensors to observe and record data, thus deepening their understanding of the condition of the equipment. Among the most critical assets that require such vigilance are power transformers, the failure of which may result in extensive operational interruptions and substantial financial repercussions.

A power transformer represents a highly complex and technically advanced system that can experience internal or external faults [1]. Global statistics indicate that a substantial majority of transformer faults (approximately 70% to 80%) are attributed to internal factors [2]. Some common internal faults include turn-to-turn fault, core lamination breakdown, and bushing failure. External faults, on the other hand, include deformation or failure, line-to-line or line-to-ground faults, overloading, and under/overfrequency [1]. Internal faults often require ad-

vanced diagnostic techniques to detect early-stage malfunction detection, while external faults may require external protection system or various robust system-level safeguards to mitigate cascading failures.

Fault diagnosis in power transformers commonly employs methods including frequency response analysis (FRA), dissolved gas analysis (DGA), and partial discharge (PD) analysis. DGA monitors gases such as hydrogen and methane in transformer oil to detect thermal and electrical faults, such as arcing and overheating [2]–[3]. That said, the accuracy of DGA methods is influenced by boundary issues, unresolved diagnostic codes, and the complexity of multi-fault scenarios. Furthermore, DGA data alone may provide an incomplete representation of transformer condition and typically must be integrated with additional monitoring data to ensure more accurate fault diagnosis [4]. FRA, in contrast, is employed to identify mechanical and electrical faults in transformers by examining the system's frequency response characteristics [5]. In (off-line) FRA, typically the system is first de-energized and then low-voltage sinusoidal signals covering a wide range of frequencies are injected into one terminal and the response is measured from another to obtain the transfer function. However, the lack of FRA result analysis procedures that compare the obtained transferred function with

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a fingerprint sometimes can lead to subjective evaluations based on the engineer's experience in interpreting the data [6].

The PD technique [7] identifies insulation degradation by capturing high-frequency electrical pulses. However, despite its high sensitivity, the analysis and processing of the characteristic parameters of the partial discharge source for fault diagnosis can be complex [8].

Infrared thermography has also been utilized to detect abnormal thermal patterns on transformer surfaces by capturing temperature distributions, enabling the identification of localized hot spots associated with insulation degradation, overloading, or cooling system malfunctions [9]. Similarly, acoustic-based monitoring techniques analyze sound or vibration signals generated by transformer operation to identify anomalies related to partial discharges, mechanical deformation, or core magnetostriction effects [10,11].

In recent years, however, the analysis of vibration patterns of the inner parts of the transformer such as the core and the windings has become an effective diagnostic technique for the prognosis of transformer faults [12]. This is a viable solution because the core and the winding vibrations are directly related to linkage and leakage fluxes, which are themselves functions of both the physical structure of the inner parts as well as the electrical properties of the system [13]. However, the degree to which the vibration signal is useful depends on how effectively it can be analyzed. This is precisely where recent advances in machine learning (ML) become very useful to interpret complex vibration signals, which is our primary concern in this work. Nonetheless, ML techniques have already been used not only to analyze these signals but also for an improved analysis of the data collected across DGA, FRA, and PD techniques.

From an ML perspective, the transformer fault diagnosis using vibration signals problem is often formulated as a supervised learning task with two key stages: (i) feature extraction from collected time series data containing information about the transformer condition; and (ii) constructing a predictive model using the extracted features. In [14], the authors used a combination of k-nearest neighbors and decision trees to enhance the precision of DGA in transformer fault detection. In [15], an improved fault diagnosis was proposed using support vector machines to analyze and classify data obtained from DGA.

Another well-known ML algorithm is Artificial Neural Network (ANN). Enabled by advances in parallel computing hardware and algorithmic improvements, multilayered ANNs have achieved breakthroughs in various applications, including image classification [16]–[17], natural language processing (NLP) [18]–[19], and speech recognition [20].

Insofar as power systems are concerned, ANNs have been used to forecast faults in power systems [21] and pinpoint faults in power transmission lines [22]. In [12], we used a special type of ANN known as multilayer perceptron (MLP) to forecast transformer over/under excitation voltage and inter-turn short-circuit fault. Transformer vibration signals were mathematically characterized as a composite of sinusoidal functions with undetermined parameters. The estimated parameters of these sinusoids (i.e., extracted features) were then used to train our predictive models. In a follow-up study [23], we relied on recurrent neural networks (RNNs), including a gated recurrent unit (GRU) and a long short-term memory (LSTM) models, to predict the aforementioned faults in transformer. As a result, in that study we did not use any explicit feature extraction mechanism such as sum of sinusoids that was used previously in [12].

Following the introduction of the Transformer architecture by [24] for sequence-to-sequence tasks, its foundational mechanism—*scaled dot-product self-attention*—has garnered substantial research interest. Unlike RNN-based architectures that process data sequentially, self-attention mechanisms enable parallel processing of all input sequence elements [25]. The self-attention operation maps each input element to a *context* vector, derived as a weighted aggregation based on its compatibility with all other sequence elements. This capacity for contextual encoding establishes self-attention as a particularly powerful approach

for sequence modeling [26]. Nevertheless, employing multiple parallel self-attention operations, termed multi-head self-attention (MHSA), typically yields superior performance compared to single-head attention. MHSA facilitates the evaluation of element-wise compatibility across diverse learned representational subspaces [24]. The effectiveness of MHSA has been demonstrated across numerous domains, including computer vision [27], [28], [29], temporal forecasting [30], and speech processing [20]–[31]. In the field of remote sensing, for instance, Mei et al. [32] proposed a Group-Aware Hierarchical Transformer (GAHT) for hyperspectral image classification, where MHSA serves as the core feature extractor to capture global spatial–spectral dependencies across long-range positions. Song et al. [33] further proposed RFormer+, a rotation-invariant transformer that leverages self-attention mechanisms to learn robust features under arbitrary rotations in remote sensing imagery.

These observations motivate the present study in which we aim to examine the utility of MHSA to enhance the performance of our best RNN models in [23] to predict transformer fault-related indicators from vibration signals. Specifically, we employ MHSA in conjunction with bidirectional gated recurrent units (BiGRU) and bidirectional long short-term memory networks (BiLSTM). These architectures belong to the gated RNN family, specifically designed to overcome the vanishing and exploding gradient problems of vanilla RNNs [34], enabling more effective capture of temporal dependencies in sequential data. These architectures were selected for the current study for two reasons: (1) they were the best-performing models in our previous study [23], and (2) their prior results on the same dataset serve as a natural benchmark, allowing us to directly measure the performance gains achieved by augmenting BiGRU and BiLSTM with MHSA. This hybrid approach has been examined in other application domains as well. In particular, [35] and [36] showed that using MHSA in RNNs can increase the performance of these models in text sentiment classification task. Unlike prognostic approaches that focus on predicting future degradation or remaining useful life, this work aims at the prediction of fault-related diagnostic indicators from vibration signals, which can support subsequent fault detection and diagnosis.

The remainder of this paper is organized as follows. In Section 2, we mathematically represent the scaled dot-product self-attention as well as the recurrent neural networks utilized in our study. Section 3 describes the experimental setup and the dataset collected for our study. Section 4 discusses model training, selection, and evaluation. Empirical results comparing the performance of BiLSTM and BiGRU with MHSA-BiLSTM and MHSA-BiGRU are also presented in Section 4. In Section 5, we summarize our findings and highlight the contribution of the attention mechanism in our application. Finally, Section 6 concludes the paper.

2. Multi-head self-attention and recurrent neural networks

In this work, we use MHSA-based BiGRU and BiLSTM hybrid models to predict transformer faults using vibration time series. In this section, first, the structure and functioning of RNNs and their variants are described, followed by a detailed explanation of the constructed hybrid model.

2.1. Recurrent neural networks

RNNs fundamentally differ from feed-forward architectures in their ability to capture temporal dependencies through a set of hidden states in addition to the input observations [37, p. 415]. These hidden states preserve information from previously processed elements of the sequence. Hence, they are widely used in applications such as NLP [38], time series forecasting [39], and speech recognition [40]. Let $\mathbf{x}_t$, $\mathbf{h}_t$, and $\mathbf{y}_t$ represent the input vector of dimension $C \times 1$, the hidden state vector of dimension $P \times 1$, and the output vector of dimension $L \times 1$, respectively. Given an input sequence $(\mathbf{x}_1, \dots, \mathbf{x}_T)$, a conventional RNN

produces a sequence of hidden states ($\mathbf{h}_1, \dots, \mathbf{h}_T$) along with their associated outputs ($y_1, \dots, y_T$) through iterative application of the following recurrence relations for $t = 1, \dots, T$:

$$\mathbf{h}_t = f_h(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{W}_{xh}\mathbf{x}_t + \mathbf{b}_h) \quad (1)$$

$$y_t = \mathbf{w}_{hy}^T \mathbf{h}_t + b_y \quad (2)$$

where $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$, and $\mathbf{w}_{hy}$ represent weight parameter matrices with respective dimensionalities of $P \times C$, $P \times P$, and $P \times 1$, $\mathbf{b}_h$ and b_y denote bias vectors of corresponding dimensions, and f_h signifies an element-wise nonlinear activation function.

The main challenges in training vanilla RNNs are the vanishing and exploding gradient problems [34]. The exploding gradient problem occurs when the gradient grows uncontrollably during training, leading to numerical instability. Conversely, the vanishing gradient problem arises when gradients diminish excessively during training, preventing effective model parameter updates [41].

To alleviate these problems, LSTM networks were proposed in [42]. These networks have gate mechanisms that regulate information flow. In an LSTM, the mapping from inputs to hidden states in (1) is replaced with the following composite function:

$$\mathbf{i}_t = \sigma(\mathbf{W}_{xi}\mathbf{x}_t + \mathbf{W}_{hi}\mathbf{h}_{t-1} + \mathbf{b}_i) \quad (3)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_{xf}\mathbf{x}_t + \mathbf{W}_{hf}\mathbf{h}_{t-1} + \mathbf{b}_f) \quad (4)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_{xc}\mathbf{x}_t + \mathbf{W}_{hc}\mathbf{h}_{t-1} + \mathbf{b}_c) \quad (5)$$

$$\mathbf{c}_t = \mathbf{i}_t \odot \tilde{\mathbf{c}}_t + \mathbf{f}_t \odot \mathbf{c}_{t-1} \quad (6)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_{xo}\mathbf{x}_t + \mathbf{W}_{ho}\mathbf{h}_{t-1} + \mathbf{b}_o) \quad (7)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (8)$$

where $\odot$ represents element-wise multiplication and σ denotes the element-wise sigmoid activation function. The matrices $\mathbf{W}_{xi}$, $\mathbf{W}_{xf}$, $\mathbf{W}_{xc}$, and $\mathbf{W}_{xo}$ are of dimension $P \times C$, while $\mathbf{W}_{hi}$, $\mathbf{W}_{hf}$, $\mathbf{W}_{hc}$, and $\mathbf{W}_{ho}$ are $P \times P$ matrices. The bias vectors $\mathbf{b}_i$, $\mathbf{b}_f$, $\mathbf{b}_c$, and $\mathbf{b}_o$ each have size $P \times 1$. Furthermore, $\mathbf{i}_t$, $\mathbf{f}_t$, $\mathbf{c}_t$, and $\mathbf{o}_t$ represent the input gate, forget gate, cell state, and output gate vectors, respectively, each of dimension $P \times 1$.

The GRU, introduced by Cho et al. [18], represents a more recent and computationally efficient architecture compared to the LSTM. Similar to the LSTM architecture, the GRU utilizes gating mechanisms to control information propagation throughout the network. Within a GRU, the fundamental components consist of the reset gate ($\mathbf{r}_t$) and the update gate ($\mathbf{z}_t$). The transformation from input to hidden state in a GRU is expressed as

$$\mathbf{z}_t = \sigma(\mathbf{W}_{xz}\mathbf{x}_t + \mathbf{W}_{hz}\mathbf{h}_{t-1} + \mathbf{b}_z) \quad (9)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_{xr}\mathbf{x}_t + \mathbf{W}_{hr}\mathbf{h}_{t-1} + \mathbf{b}_r) \quad (10)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}(\mathbf{h}_{t-1} \odot \mathbf{r}_t) + \mathbf{b}_h) \quad (11)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \tilde{\mathbf{h}}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1} \quad (12)$$

where $\mathbf{W}_{xz}$, $\mathbf{W}_{xr}$, $\mathbf{W}_{xh}$, and $\mathbf{W}_{hh}$ represent weight parameter matrices of dimensions $P \times C$, $P \times C$, $P \times C$, and $P \times P$, respectively, while $\mathbf{b}_z$, $\mathbf{b}_r$, and $\mathbf{b}_h$ correspond to bias vectors of size $P \times 1$.

The aforementioned RNN-based architectures are *unidirectional* models in that they process a given sequential data in one direction (from a lower time index t to a higher one). Bidirectional RNNs, however, augment the functionality of unidirectional architectures by simultaneously processing input sequences in both temporal directions [43]. This architecture comprises two parallel layers: a forward layer that traverses the sequence from initial to final time steps, and a backward layer that processes the sequence in reverse chronological order. The final output is derived by integrating information from both forward and backward hidden state sequences. Mathematically, bidirectional RNNs are formally expressed as

$$\bar{\mathbf{h}}_t = f_h(\bar{\mathbf{W}}_{xh}\mathbf{x}_t + \bar{\mathbf{W}}_{hh}\bar{\mathbf{h}}_{t-1} + \bar{\mathbf{b}}_h) \quad (13)$$

$$\bar{\mathbf{h}}_t = f_h(\bar{\mathbf{W}}_{xh}\mathbf{x}_t + \bar{\mathbf{W}}_{hh}\bar{\mathbf{h}}_{t+1} + \bar{\mathbf{b}}_h) \quad (14)$$

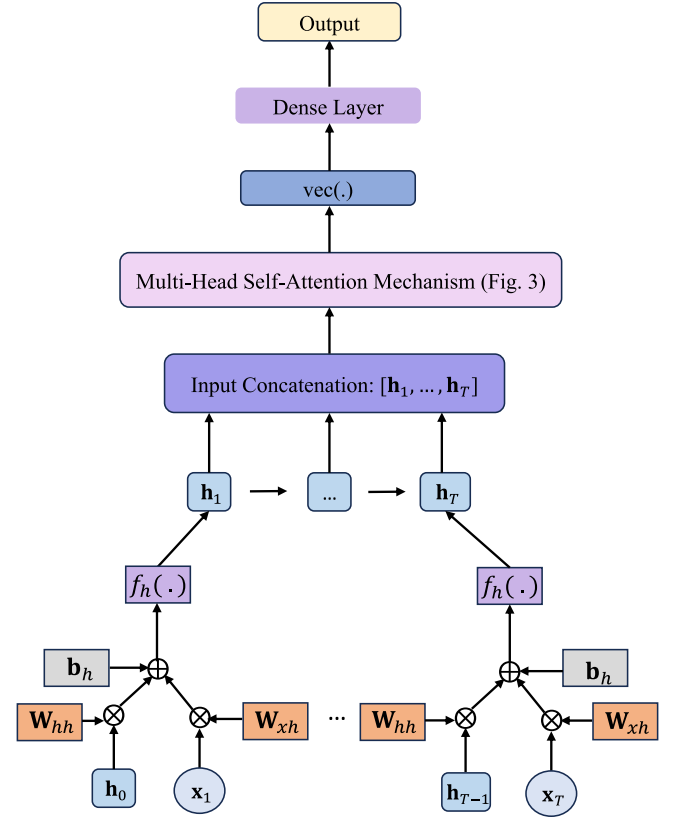


Fig. 1. The multi-head self-attention-based rnn model.

$$y_t = \bar{\mathbf{w}}_{hy}^T \bar{\mathbf{h}}_t + \bar{\mathbf{w}}_{hy}^T \mathbf{h}_t + b_y \quad (15)$$

where $\bar{\mathbf{W}}_{xh}$, $\bar{\mathbf{W}}_{hh}$, $\bar{\mathbf{W}}_{xh}$, $\bar{\mathbf{W}}_{hh}$, $\bar{\mathbf{w}}_{hy}$, $\bar{\mathbf{b}}_h$, and b_y are weight matrices or bias terms of appropriate dimensions. BiLSTM and BiGRU are formed by replacing $\bar{\mathbf{h}}_t$ and $\mathbf{h}_t$ with the composite functions presented in (3)-(8) and (9)-(12), respectively. These architectures can incorporate multiple hidden layers by treating the hidden state vector of one layer as input to the next [44].

2.2. Multi-head attention-based RNN

This investigation aims to examine the benefit of integrating MHSA with RNNs. In this regard, the hidden states of the last layer of RNNs (BiGRU and BiLSTM) are used to form a single matrix $\mathbf{X} = [\mathbf{h}_1, \dots, \mathbf{h}_T] \in \mathbb{R}^{P \times T}$, which is fed to the MHSA layer. The flattened MHSA output is then utilized as the input to a dense layer that makes the final prediction. Fig. 1 presents an overview of the MHSA-based hybrid approach.

Several attention mechanisms have been proposed in the literature for augmenting RNN-based architectures. The additive attention mechanism proposed by Bahdanau et al. [45] computes a context vector as a weighted sum of encoder hidden states to guide a decoder at each decoding step. Similarly, Luong et al. [46] proposed multiplicative attention variants for sequence-to-sequence tasks. However, both of these mechanisms were designed specifically for encoder-decoder architectures targeting sequence-to-sequence tasks such as machine translation, and are therefore not directly applicable to our many-to-one regression formulation, where a single continuous fault indicator must be predicted from an input vibration sequence. Single-head self-attention represents another alternative; however, it attends to sequence elements using a single learned projection, limiting its capacity to capture compatibility across diverse representational subspaces simultaneously. In contrast, MHSA [24] employs multiple parallel attention heads, each operating in a different learned subspace, enabling richer and more expressive

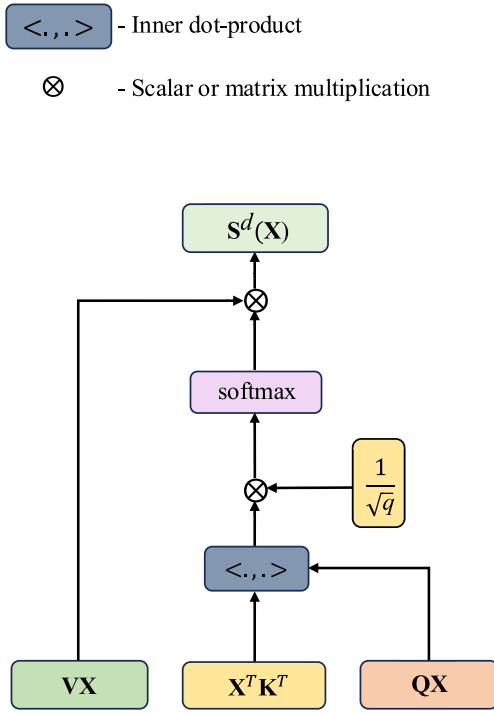


Fig. 2. The scaled dot-product self-attention.

temporal dependency modeling from the hidden state sequence produced by the final recurrent layer. For these reasons, MHSA was selected as the most appropriate attention mechanism for augmenting the BiGRU and BiLSTM models in the proposed fault prognosis framework.

2.2.1. The scaled dot-product self-attention

Given an input sequence $\mathbf{X} \in \mathbb{R}^{P \times T}$, the scaled dot-product self-attention, denoted as $\mathbf{S}^d(\mathbf{X})$, is a mapping from $\mathbb{R}^{P \times T} \rightarrow \mathbb{R}^{d \times T}$, which is characterized by

$$\mathbf{S}^d(\mathbf{X}) = \mathbf{VX} \times \text{softmax}\left(\frac{\mathbf{X}^T \mathbf{K}^T \mathbf{QX}}{\sqrt{q}}\right) \quad (16)$$

where $\mathbf{V} \in \mathbb{R}^{d \times P}$, $\mathbf{K} \in \mathbb{R}^{q \times P}$, and $\mathbf{Q} \in \mathbb{R}^{q \times P}$ are projection matrices learned during training, $\text{softmax}(\mathbf{A})$ is applied to each column of matrix $\mathbf{A}$, q represents the attention dimensionality—a hyperparameter subject to tuning—and d corresponds to the dimensionality of the output matrix columns (the *context* vectors). Fig. 2 shows a graphical representation of this operation.

2.2.2. Multi-head self-attention

It is generally acknowledged that having multiple self-attention operations implemented in parallel helps in jointly learning the compatibility of sequence elements using different learned projections. The multi-head self-attention used in [24] is a mapping $\text{MHSA}^{d_k}(\mathbf{X}): \mathbb{R}^{P \times T} \rightarrow \mathbb{R}^{d_k \times T}$ that uses k self-attention heads. In particular, we have

$$\text{MHSA}^{d_k}(\mathbf{X}) = \mathbf{W}[\mathbf{S}_1^d(\mathbf{X})^T, \dots, \mathbf{S}_k^d(\mathbf{X})^T]^T \quad (17)$$

where $\mathbf{W} \in \mathbb{R}^{d_k \times kd}$ denotes a trainable projection matrix and d_k specifies the column dimensionality in the output of the $\text{MHSA}^{d_k}(\mathbf{X})$ operation. Each component $\mathbf{S}_i^d(\mathbf{X})$, $i = 1, \dots, k$ corresponds to the output produced by the i -th self-attention head as defined in (16). The MHSA operation described by (17) is illustrated in Fig. 3.

2.3. Dense layer

Given an input sequence of length T , the model computes the output of $\text{MHSA}^{d_k}(\mathbf{X})$ and produces a single prediction, denoted as y . In

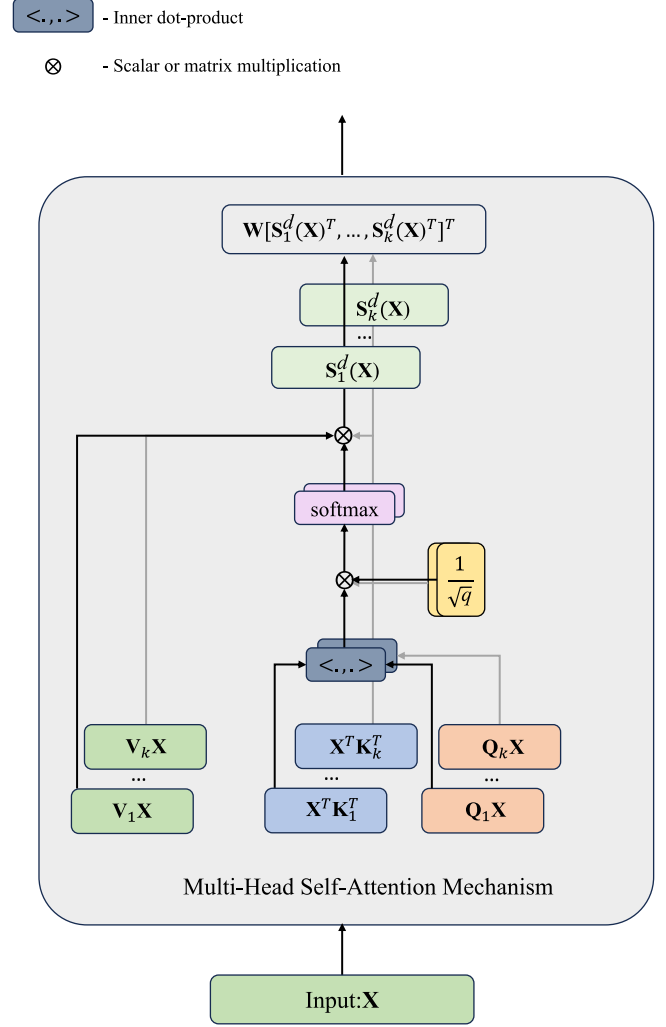


Fig. 3. Multi-head self-attention mechanism.

particular, y is computed by first vectorizing (flattening) the output of $\text{MHSA}^{d_k}(\mathbf{X})$, which is then used as input to a dense layer with no activation function; that is to say,

$$y = \mathbf{w}^T g(\mathbf{X}) + b \quad (18)$$

where $g(\mathbf{X}) \triangleq \text{vec}(\text{MHSA}^{d_k}(\mathbf{X}))$, $\text{vec}(\mathbf{A})$ denotes vectorization operation (i.e., stacking columns of matrix $\mathbf{A}$ in one column vector), $\mathbf{w} \in \mathbb{R}^{d_k \times T \times 1}$ is a learnable weight vector, and $b \in \mathbb{R}$ is a scalar bias term.

Fig. 4 presents the overall framework of the implemented approach. The raw vibration signal acquired from the transformer is first segmented into fixed-length windows of 0.02 seconds ($T = 60$ samples at a sampling frequency of $F_s = 3$ kHz) and normalized using the min-max normalization computed from the training set. The preprocessed segments are then fed into a stacked bidirectional recurrent neural network, either BiGRU or BiLSTM, which processes the input sequence in both temporal directions across multiple layers to produce a sequence of hidden state vectors. These hidden states are subsequently passed to a Multi-Head Self-Attention (MHSA) module, which models interdependencies across all time steps to produce an enriched representation of the input sequence. The MHSA output is then flattened into a single vector and passed through a fully connected dense layer to produce the final scalar prediction.

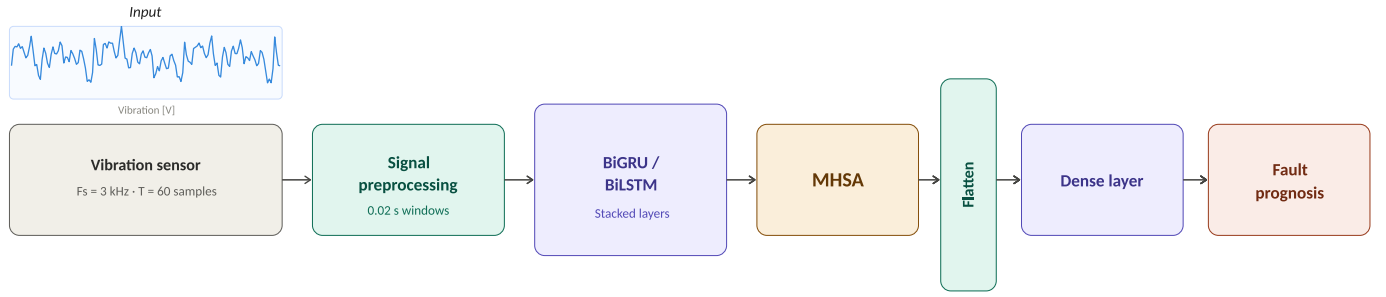


Fig. 4. Overall framework of the vibration-based transformer fault prognosis approach using MHSA-augmented bidirectional recurrent neural networks. The raw vibration signal is segmented and normalized before being processed by stacked BiGRU or BiLSTM layers, followed by a Multi-Head Self-Attention (MHSA) module, a flattening operation, and a fully connected dense layer to produce the predicted fault indicator.

3. Experimental setup

3.1. Dataset

In this work, excitation voltage deviation and inter-turn short-circuit current were selected as fault-related indicators because they can be introduced in a controlled, repeatable, and non-destructive manner on a commercial transformer, enabling systematic experimentation and reliable data labeling. These indicators correspond to underexcitation and overexcitation conditions and inter-turn short-circuit faults, respectively. In contrast, many other fault types, such as insulation degradation, winding deformation, or core structure damage, are irreversible and destructive, making them unsuitable for controlled laboratory testing. In addition, these selected faults allow precise control of fault severity, providing a well-defined basis for evaluating the learning behavior of data-driven models.

3.1.1. Transformer underexcitation/overexcitation voltage

Transformer underexcitation and overexcitation are abnormal operating conditions in distribution transformers that may lead to distorted waveforms, harmonic distortion, increased core losses, and thermal stress. In particular, transformer overexcitation can cause saturation of the transformer core, flux deviation, and eventually secondary current distortion. Conversely, underexcitation may elevate core losses through fundamental frequency flux modifications, diminish output voltage levels, and amplify copper losses.

To simulate transformer underexcitation and overexcitation conditions, vibration analysis was conducted during a preliminary study on a three-phase, double-winding, 15 kVA, 0.4/0.4 kV cast-resin dry-type isolation transformer (see Fig. 6) [12]. The primary-side excitation voltage of the transformer was systematically varied from 320 V to 440 V (80% to 110% of the rated value) in 20 V steps (5% increments) using a three-phase autotransformer. Vibration measurements were acquired by sensors installed on the core and at the top of the windings, enabling the simultaneous capture of vibration signals associated with both core and winding under different excitation conditions. The vibration measurements were obtained from controlled laboratory experiments rather than field measurements; however, the tested transformer represents a typical distribution transformer configuration, allowing the collected vibration data to reflect behavior relevant to practical industrial applications. The recorded vibration signals had durations ranging from 28.6 s to 48.1 s depending on the operating condition and were segmented into 0.02 s intervals, each containing 60 observations (corresponding to a sampling rate of 3 kHz). This segmentation resulted in 1430 to 2405 samples per operating condition, which were used as input data for model training and validation.

Fig. 5 illustrates representative vibration signal segments recorded under three distinct excitation voltage conditions: underexcitation (320 V, 80% of rated), nominal (400 V, 100% of rated), and overexcitation (440 V, 110% of rated). To enable a fair visual comparison, all subplots

share the same vertical axis range. As can be observed, increasing the excitation voltage leads to a progressive growth in vibration amplitude – the underexcitation case (320 V) exhibits a relatively narrow amplitude band, while the overexcitation case (440 V) shows substantially larger oscillations. These differences, while physically consistent, are subtle in the raw signal domain, which motivates the use of deep learning-based approaches capable of automatically extracting discriminative temporal features for accurate fault prognosis.

Accurate prediction of excitation voltage deviation is directly relevant to transformer fault diagnosis because abnormal excitation levels modify the magnetic flux density in the core, leading to increased magnetostriction forces and altered vibration patterns. Therefore, estimating excitation voltage from vibration measurements provides an indirect yet physically grounded indicator of abnormal operating conditions, enabling early identification of faults such as core saturation, excessive fluxing, and associated thermal stress.

3.1.2. Interturn short-circuit current

Another test object was used for interturn short-circuit analysis. To emulate a transformer interturn short-circuit fault while the transformer was in service (under load), we used a 240/30/30 V, 252 VA single-phase transformer. We collected vibration data under various load conditions. Interturn short-circuit faults were simulated by connecting a variable resistor across two winding turns on a disk, with fault initiation and clearance controlled via a miniature circuit breaker. This process created a turn-to-turn fault, simulating an interturn short circuit. The severity of the fault was varied by adjusting the resistor and altering the circulating short-circuit current.

Fig. 7 presents representative vibration signal segments recorded under three different interturn short-circuit current conditions (8 A, 10 A, and 12 A). As can be observed, the raw vibration signals exhibit subtle variations across the three current levels, with higher short-circuit currents introducing slight changes in signal amplitude and waveform morphology. The inherently subtle nature of these differences further underscores the necessity of employing deep learning models capable of automatically extracting discriminative temporal features for accurate interturn short-circuit fault prognosis.

The vibration time series data were segmented into 0.02-second intervals. This process subdivides each second of the collected vibration waveforms into 50 distinct sequences (hereafter referred to as observations). These observations were used for models training and evaluation. With a sampling frequency of 3 kHz, each observation comprised 60 consecutive data points extracted from the original temporal sequence (thus, $T = 60$). The dataset underwent random permutation, and 25% of the complete observations corresponding to each voltage injection level were reserved as a validation subset, employed for model optimization (hyperparameter adjustment). The tables detailing observation durations and various loading conditions for transformer overexcitation and load current cases, respectively, are presented in our previous publication [23].

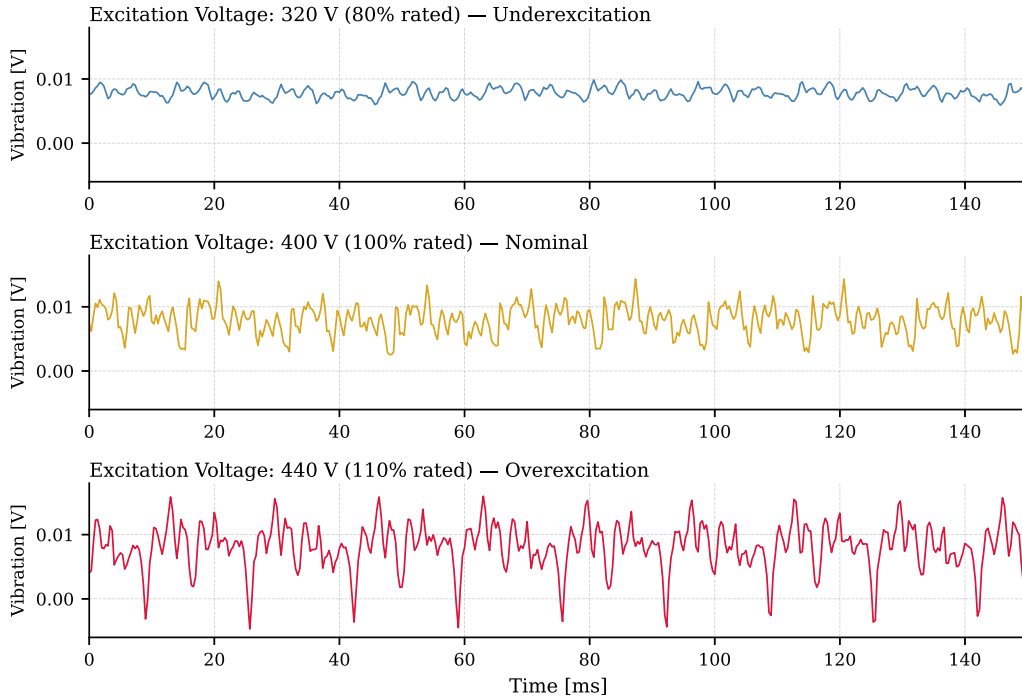


Fig. 5. Representative vibration signal segments recorded under underexcitation (320 V, 80% of rated), nominal (400 V, 100% of rated), and overexcitation (440 V, 110% of rated) conditions. All subplots share the same vertical axis range for direct comparison.

Table 1
Hyperparameter space for mhsa-bigr and mhsa-bilstm models.

Model	MHSA		RNN		Batch Size	Learning rate	Optimizer	Epochs
	k	q	M	N				
MHSA-BiGRU	1, 2, 4	16, 32, 64	32, 64, 128	3, 5	64	10^{-3}	AdamW	1 to 150
MHSA-BiLSTM	1, 2, 4	16, 32, 64	32, 64, 128	3, 5	64	10^{-3}	AdamW	1 to 150

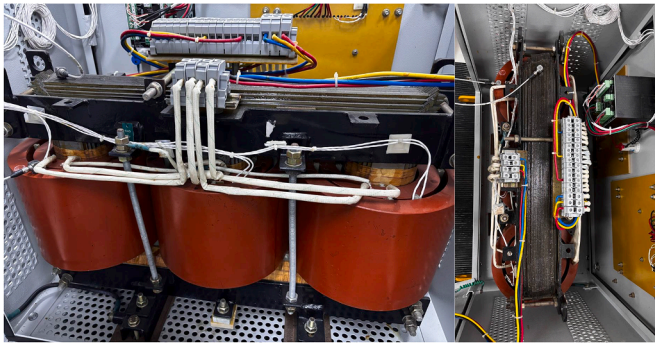


Fig. 6. A 0.4/0.4 kV three-phase cast-resin transformer instrumented with vibration transducers and a data acquisition system.

4. Model construction, selection, and evaluation

In [23], we observed that BiGRU and BiLSTM were the best performing models for predicting under/overexcitation voltage and interturn short-circuit current, respectively. As a result, in this study, we stack MHSA on top of BiGRU and BiLSTM to predict the aforementioned faults. Nonetheless, we did not use the values of hyperparameters identified in [23] for BiGRU and BiLSTM, and performed hyperparameter tuning afresh for each case; that is to say, MHSA-BiGRU and MHSA-BiLSTM models were trained with different configurations to select the

best-performing model for predicting the faults. The source code for all experimental implementations is publicly available.¹

In particular, models with varying numbers of hidden units $M = 32, 64, \text{ and } 128$, and numbers of layers $N = 3 \text{ and } 5$ were trained and evaluated on case-specific validation sets.

Table 1 presents the full description of the searched hyperparameter space. The hyperparameter choices specified in Table 1 result in 3 (number of heads) $\times$ 3 (attention dimensionality) $\times$ 3 (number of hidden units) $\times$ 2 (number of layers) $\times$ 1 (batch size) $\times$ 1 (learning rate) $\times$ 1 (optimizer: AdamW [47]) $\times$ 150 (number of epochs) $\times$ 2 = 8100 trained models to select from for each fault category: underexcitation/overexcitation voltage faults and interturn short-circuit faults.

For each experimental case, model selection was based on minimizing MSE performance on the validation dataset. Specifically, the MSE metric is computed as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (pv_i - av_i)^2 \quad (19)$$

where n denotes the total number of test samples, and pv_i and av_i represent the estimated and actual values of the response variable for the i -th observation, respectively. Fig. 9 shows $-\log_{10}(\text{MSE})$ at the best-performing epoch for the 54 models defined by the hyperparameter combinations (k, q, M , and N) outlined in Table 1. Interestingly, as can be seen in Fig. 9, for both MHSA-BiGRU (used for excitation voltage prediction) and MHSA-BiLSTM (used for interturn short-circuit prediction) the best-performing model is obtained for the same values of hy-

¹ https://github.com/assanalimarwa/Transformer_Fault_prognosis

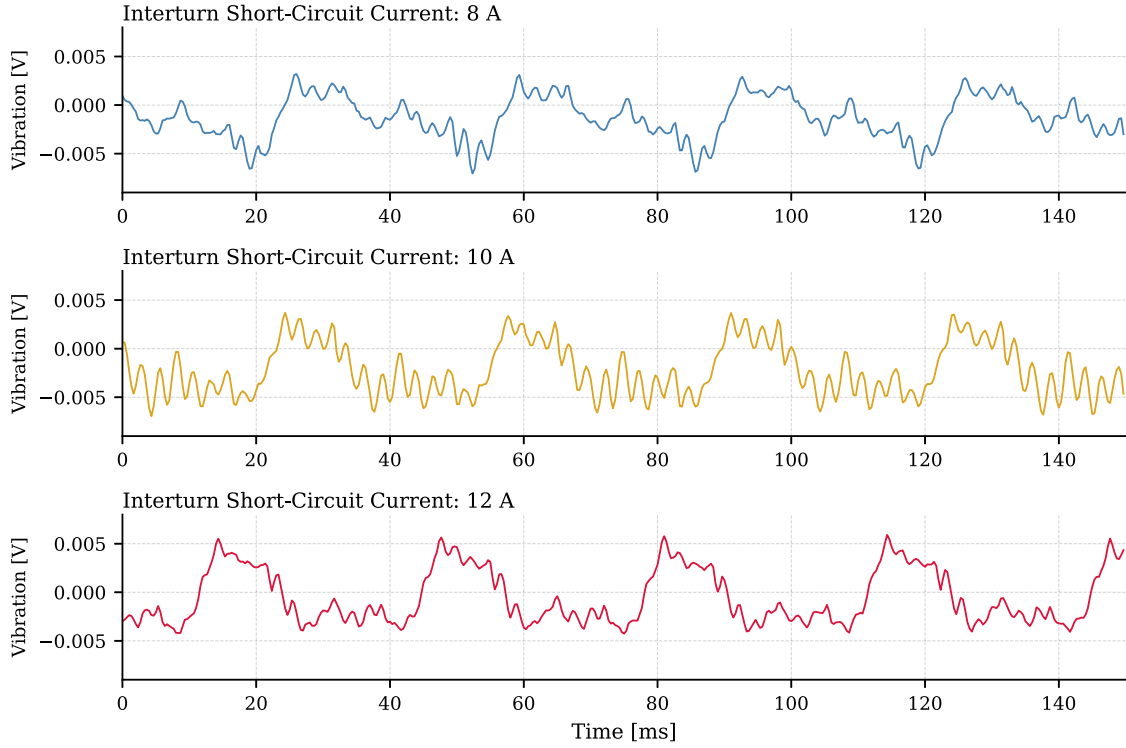


Fig. 7. Representative vibration signal segments recorded under different interturn short-circuit current conditions (8 A, 10 A, and 12 A). All subplots share the same vertical axis range for direct comparison.

Table 2

Performance of the best MHSA-BiGRU and MHSA-BiLSTM models on a validation set.

Models	MHSA		RNN		MSE	Total Parameters
	k	q	M	N		
MHSA-BiGRU	1	16	32	5	7.80×10^{-7}	90,033
MHSA-BiLSTM	1	16	32	5	8.41×10^{-4}	116,081

perparameters: $k = 1$, $q = 16$, $M = 32$, and $N = 5$. Table 2 presents the specifications and validation set MSE of these models.

To evaluate our selected models and compare their performance with the best-performing BiGRU (for excitation voltage) and BiLSTM (for interturn short-circuit) from our previous study [23], we tested them on the same datasets as those in [23]. Similar to [23], RAE and RRSE performance metrics were used to facilitate comparison and, at the same time, provide a more intuitive understanding of the predictive capacity of our constructed models. In particular,

$$\text{RAE} = 100 \times \frac{\sum_{i=1}^n |pv_i - av_i|}{\sum_{i=1}^n |av_i - \bar{av}|} \quad (20)$$

$$\text{RRSE} = 100 \times \sqrt{\frac{\sum_{i=1}^n (pv_i - av_i)^2}{\sum_{i=1}^n (av_i - \bar{av})^2}} \quad (21)$$

where $\bar{av}$ denotes the average of the actual values for the target variable. These two metrics evaluate predictive model performance against a baseline constant predictor $\bar{av}$. In particular, when a predictive model consistently outputs $\bar{av}$ for all target observations, RAE and RRSE both attain values of 100%, signifying a total lack of predictive power. Conversely, under ideal predictive conditions where $pv_i = av_i, \forall i$, both metrics converge to zero.

Table 3 presents the computational complexity of the plain and MHSA-augmented models in terms of total parameter count and float-

Table 3

Computational complexity comparison between plain and MHSA-augmented models.

Models	Parameters	FLOPs (K)
BiGRU	82,049	99.8
MHSA-BiGRU	90,033	627.2
BiLSTM	108,097	134.6
MHSA-BiLSTM	116,081	662.0

ing point operations (FLOPs). The MHSA integration adds only 7,984 parameters to both BiGRU and BiLSTM, corresponding to an overhead of 9.7% and 7.4%, respectively. While the FLOPs increase more notably due to the quadratic scaling of the attention mechanism with respect to sequence length, the absolute values remain modest given the short input sequences ($T = 60$) used in this study.

As can be seen in Table 3, the parameter overhead introduced by MHSA is minimal in both cases. The relatively higher FLOPs compared to the plain models is attributable to the scaled dot-product attention operations, which scale as $\mathcal{O}(T^2 \cdot d)$, where $T = 60$ is the sequence length and $d = 16$ is the attention dimensionality. Despite this increase, the absolute FLOPs remain small, confirming that MHSA can be integrated into the RNN architecture without introducing prohibitive computational burden.

Fig. 8 shows the test-set RAE and RRSE of the constructed MHSA-BiGRU and MHSA-BiLSTM models for excitation voltage and interturn short-circuit fault prediction, respectively. For comparison purposes, we also show the same performance metrics of the best RNN models constructed in our previous study [23]. As can be seen in this figure, the use of MHSA improved the performance of both BiGRU and BiLSTM in both cases and in terms of both performance metrics.

In particular, the constructed MHSA-BiGRU ($k = 1$, $q = 16$, $M = 32$, and $N = 5$) for excitation voltage prediction achieved an RAE of 0.29%, which compared with 0.56% RAE of the BiGRU constructed in [23]

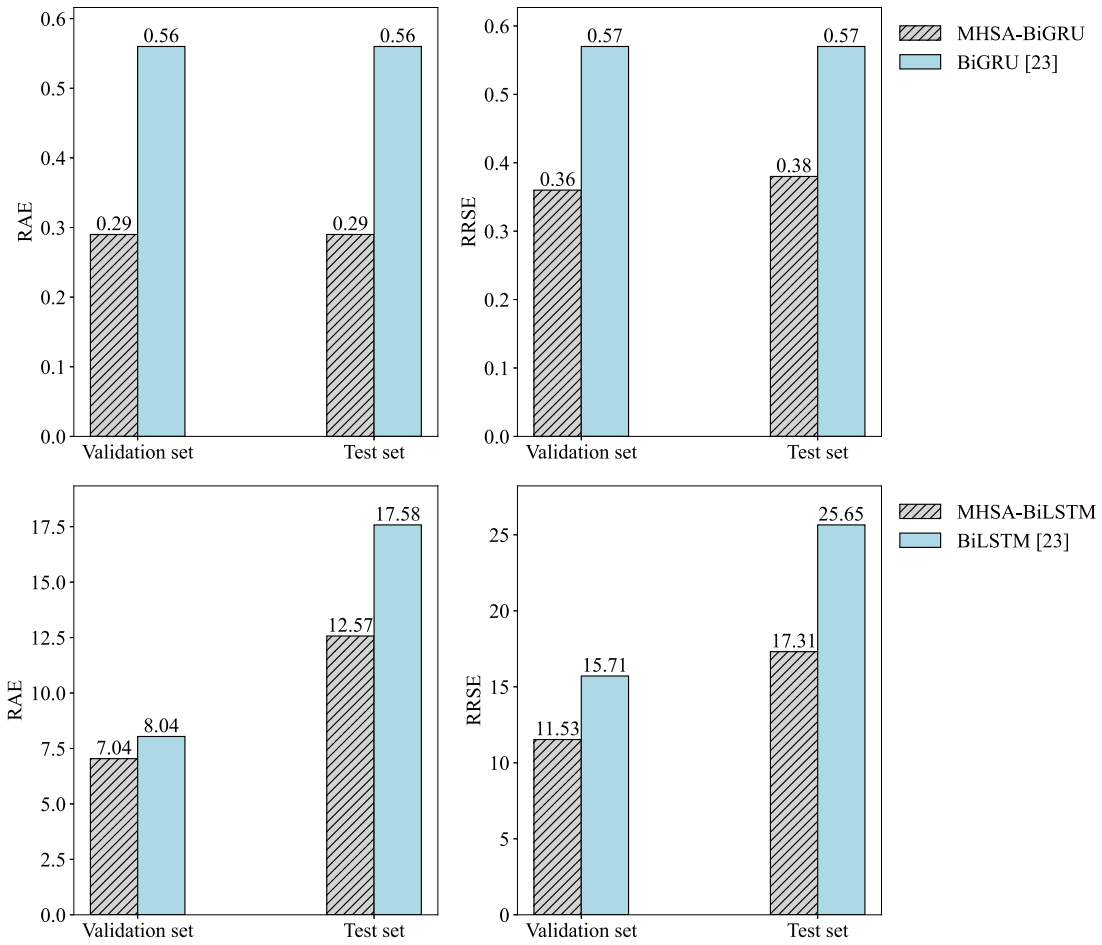


Fig. 8. The comparison of RAE and RRSE results between the MHSA-BiGRU and MHSA-BiLSTM with the simple BiLSTM and BiGRU respectively.

is a relative improvement of $\frac{0.56-0.29}{0.29} \times 100 \approx 48\%$. For interturn short-circuit fault current prediction, the constructed MHSA-BiLSTM ($k = 1$, $q = 16$, $M = 32$, and $N = 5$) achieved an RAE of 12.57%, which compared with 17.58% RAE of the BiLSTM constructed in [23] is a relative improvement of about 28%.

5. Discussion

In our initial study in this line of work [12], we utilized conventional machine learning techniques with explicit feature extraction to analyze transformer vibration signals. Following our analytical methodology for transformer vibration signal modeling, feature extraction (signal modeling) was performed using a summation of sinusoidal components with undetermined parameters encompassing phases, amplitudes, and frequencies. These unknown parameters were then estimated over each segment of the collected vibration waveforms. In the field of statistical signal processing, this technique, namely, representing time-series segments via aggregated sinusoids with unknown parametric values, is termed the segmented Prony approach for spectral line estimation [48].

Building upon this work, in [23] we explored the capabilities of deep neural networks (specifically RNNs) to predict excitation voltage and interturn short-circuit current using vibration signals. This emphasis is justified given that deep neural networks are well-suited for incorporating feature extraction within the model development process. In particular, we relied on LSTM, BiLSTM, GRU, and BiGRU models to extract informative features of vibration signals for prediction. The similar performance outcomes of the top-performing models—specifically BiGRU and BiLSTM—relative to the results obtained in [12] validated these RNNs'

capacity to identify latent patterns relevant to transformer fault prediction.

The present study investigates the efficacy of incorporating the MHSA mechanism to enhance BiGRU and BiLSTM performance in transformer fault prediction based on vibration signal analysis. This research focus is well-founded, given that self-attention mechanisms facilitate simultaneous consideration of all elements within the input sequence [25]. Specifically, self-attention exploits sequential similarity across all temporal steps to construct an enriched representation (context vector) of the input sequence. This refined representation, fundamentally derived from the compatibility relationships among all sequence elements, enables the model to selectively emphasize the most salient components of the input during prediction generation.

To assess the computational overhead introduced by the MHSA module, we analyze the theoretical complexity of each component. The dominant computational cost of a BiGRU with N stacked bidirectional layers, M hidden units, and input sequence of length T is given by $\mathcal{O}(2 \cdot N \cdot T \cdot 3 \cdot M^2)$, reflecting the three gating operations of the GRU cell processed in both temporal directions across all layers. Similarly, the BiLSTM complexity is $\mathcal{O}(2 \cdot N \cdot T \cdot 4 \cdot M^2)$, where the additional term accounts for the four gate operations of the LSTM cell. The MHSA module, operating on the hidden state sequence of length T with k attention heads and attention dimensionality q , introduces an additional complexity of $\mathcal{O}(k \cdot T^2 \cdot q)$ due to the scaled dot-product attention computation. For the best-performing model configuration identified in this study ($N = 5$, $M = 32$, $k = 1$, $q = 16$, $T = 60$), the theoretical complexity of the MHSA module amounts to $1 \times 60^2 \times 16 = 57,600$ operations, compared to $2 \times 5 \times 60 \times 3 \times 32^2 = 1,843,200$ for BiGRU and

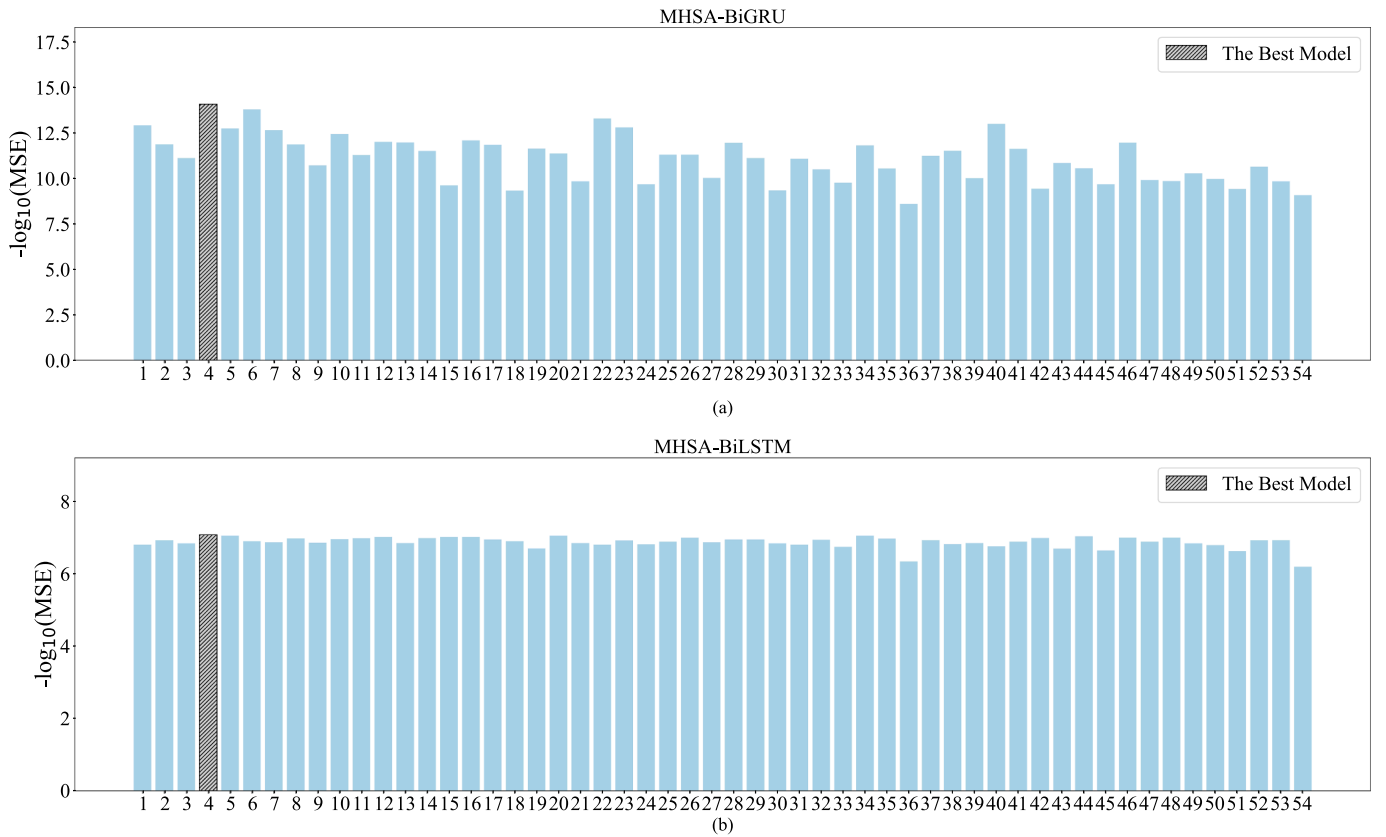


Fig. 9. The performance of (a) MHA-BiGRU and (b) MHA-BiLSTM models on a validation set in terms of $-\log_{10}(\text{MSE})$ (the higher, the better). The configuration of each model identified by the model ID on the horizontal axis is presented in Table 4.

$2 \times 5 \times 60 \times 4 \times 32^2 = 2,457,600$ for BiLSTM. This corresponds to a theoretical overhead of approximately 3.1% relative to BiGRU and 2.3% relative to BiLSTM, confirming that the MHA integration introduces negligible additional complexity. These findings are further supported by the empirical parameter counts and FLOPs reported in Table 3, where the MHA module adds only 7,984 parameters – corresponding to 9.7% and 7.4% overhead for MHA-BiGRU and MHA-BiLSTM respectively.

In our previous study [23], the employed RNN-based approach exclusively used the hidden state vector from the topmost recurrent layer at the terminal time step $t = T$. More precisely, for the many-to-one sequence mapping problem addressed, the predicted output y_T was determined as $y_T = \mathbf{w}_T^T \mathbf{h}_T^N + b_y$, in which N designates the total layer count and $\mathbf{h}_t^j$ indicates the hidden state of layer j at time instant t . Nevertheless, due to the recurrent connections in RNNs, it is generally acknowledged that $\mathbf{h}_T^N$ contains all the relevant past information in the sequence that are useful to produce the output (insofar as the RNN is able to capture/keep them). In the present study, we integrate the MHA with the RNN architecture by positioning the MHA layer subsequent to the final recurrent layer. Specifically, the MHA module receives as input the sequence of hidden state vectors generated by the terminal RNN layer. Consequently, the generation of y_T leverages not only the complete set of hidden states $\mathbf{h}_1^N, \dots, \mathbf{h}_T^N$ explicitly, but also their enhanced representations derived from interdependency modeling.

Our study has demonstrated that integrating MHA with RNNs can potentially enhance the capability of RNN to capture richer patterns in vibration data. A major limitation in our work, however, is the use of dot-product self-attention, which is highly parallelizable due to optimized matrix multiplication operations, with RNNs. In particular, the sequential processing of input sequences in RNNs makes the computations more difficult to parallelize. Nonetheless, here we were primarily concerned with the use of MHA along with best-performing RNN architec-

tures used in [23]. Future studies can also investigate various attention mechanisms along with other highly parallelizable architectures such as convolutional neural networks. Additionally, transformer-based model like the Autoformer, which is designed for long-term forecasting, could be investigated for its ability to capture extended dependencies [49]. Exploring these architectures, in addition to MHA-based RNNs, could provide valuable insights and potential performance improvements in our application.

The results highlight the strong relationship between transformer vibration signatures and underlying electrical conditions. Variations in excitation voltage influence core magnetostriction forces, while inter-turn short-circuit faults alter winding current distribution, both leading to distinct vibration patterns. The accurate estimation of these diagnostic indicators confirms that vibration signals effectively capture both core- and winding-related phenomena. Compared to conventional techniques such as dissolved gas analysis and partial discharge monitoring, the proposed approach offers a non-intrusive and cost-effective alternative for condition monitoring. However, the study is based on controlled laboratory data from a single transformer, and further validation under real operating conditions is required. Future work will focus on incorporating field data and extending the approach toward multi-sensor and prognostic applications.

The designed framework is not limited to the specific dataset used in this study. Since the model learns general relationships between transformer vibration signatures and fault-related diagnostic indicators, it can be applied to other transformer monitoring datasets with similar measurement modalities. In particular, the approach is well-suited for deployment on sister transformers with comparable design, rating, and operating conditions, where vibration characteristics exhibit strong similarity. For such cases, the trained model can be directly applied or further refined through light fine-tuning using a small amount of

Table 4
Model configurations.

Model ID	k	q	M	N
1	1	16	32	3
2	1	16	64	3
3	1	16	128	3
4	1	16	32	5
5	1	16	64	5
6	1	16	128	5
7	1	32	32	3
8	1	32	64	3
9	1	32	128	3
10	1	32	32	5
11	1	32	64	5
12	1	32	128	5
13	1	64	32	3
14	1	64	64	3
15	1	64	128	3
16	1	64	32	5
17	1	64	64	5
18	1	64	128	5
19	2	16	32	3
20	2	16	64	3
21	2	16	128	3
22	2	16	32	5
23	2	16	64	5
24	2	16	128	5
25	2	32	32	3
26	2	32	64	3
27	2	32	128	3
28	2	32	32	5
29	2	32	64	5
30	2	32	128	5
31	2	64	32	3
32	2	64	64	3
33	2	64	128	3
34	2	64	32	5
35	2	64	64	5
36	2	64	128	5
37	4	16	32	3
38	4	16	64	3
39	4	16	128	3
40	4	16	32	5
41	4	16	64	5
42	4	16	128	5
43	4	32	32	3
44	4	32	64	3
45	4	32	128	3
46	4	32	32	5
47	4	32	64	5
48	4	32	128	5
49	4	64	32	3
50	4	64	64	3
51	4	64	128	3
52	4	64	32	5
53	4	64	64	5
54	4	64	128	5

asset-specific vibration data. For transformers with different structural or operational characteristics, transfer learning techniques can be employed to adapt the model while preserving the learned diagnostic patterns.

6. Conclusion

Recurrent neural networks including BiGRU and BiLSTM have already shown remarkable performance in predicting transformer fault using vibration time series. This success is mainly attributed to the capability of these models to extract complex temporal patterns from time series. In an effort to push the performance of these models beyond their current level, in this study we placed the scaled dot-product self-attention mechanism after the last layer of both BiGRU and BiLSTM to predict overexcitation/underexcitation voltage and interturn short-circuit current faults, respectively. Our results show a tangible perfor-

mance improvement achieved when the aforementioned RNNs are augmented by the self-attention mechanism. We believe this state of affairs is considered to result from the capacity of the self-attention mechanism to utilize the sequence similarity between all time-steps to compute a richer representation of the hidden states of RNNs. This new representation of the hidden states before the decision-making layer can potentially allow the model to selectively emphasize the most salient features during prediction generation. Although real-world environments may introduce additional variability, the results indicate that attention-enhanced RNNs can capture physics-consistent vibration patterns relevant to transformer fault prognosis.

CRedit authorship contribution statement

Assanali Aidarkhan: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization; **Venera Nurmanova:** Writing – review & editing, Conceptualization; **Amin Zollanvari:** Writing – review & editing, Supervision, Conceptualization; **Mehdi Bagheri:** Writing – review & editing, Validation, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Mehdi Bagheri reports financial support was provided by Ministry of Science and Higher Education of the Republic of Kazakhstan. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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